Question	Answer	Marks
2(a)	sum / total momentum (of a system of bodies) is constant *or* sum / total momentum before = sum / total momentum after	**M1**
	for an isolated system / no (resultant) external force	**A1**
2(b)(i)	23×3.4 or $(23 + 160)v$ $23 \times 3.4 + (160 \times 0) = (23 + 160)v$	**C1**
	$v = 0.43\,\mathrm{m\,s^{-1}}$	**A1**
2(b)(ii)	$E_{(K)} = ½\, mv^2$	**C1**
	$E_2 / E_1 = ½\,(160+23) \times 0.43^2 / ½\,(23) \times 3.4^2$ $= 16.9 / 132.9$	**C1**
	$E_2 / E_1 = 0.13$	**A1**

Question	Answer	Marks
2(b)(iii)	The total kinetic energy after the collision is less than the total kinetic energy before the collision, so inelastic	B1
	OR The relative speed of approach is more than the relative speed of separation, so inelastic	(B1)
	OR $E_2 < E_1$ so inelastic	(B1)
2(c)(i)	(upthrust =) (160 + 23) × 9.81 = 1800 (N)	A1
2(c)(ii)	$\rho = F/gV$ $= 1800/(9.81 \times 1.80 \times 10^5 \times 10^{-6})$	C1
	$= 1.02 \times 10^3$ kg m^{-3}	A1
	OR $\rho = m/V$ $= 183/(1.80 \times 10^5 \times 10^{-6})$	(C1)
	$= 1.02 \times 10^3$ kg m^{-3}	(A1)